

NAWAF AL HUSSAIN KHONDOKAR

Platform Engineer | Cloud Architect | Backend Specialist

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Professional Summary

Versatile technology professional with diverse business experience and experience in software engineering, cloud architecture, and business operations. Proven expertise in designing and implementing **high-availability backend systems** using Javascript, TypeScript, and Node.js, with production systems handling **10K+ concurrent users**.

Demonstrated success in optimizing system performance, implementing secure authentication mechanisms, and automating infrastructure using Docker, Kubernetes, and CI/CD pipelines.

Unique combination of technical and business skills developed through diverse experiences including software engineering, private tutoring with AI integration, BPO sales management, and 5+ years of retail business management. Strong foundation in distributed systems, cloud-native solutions, and full-stack development, complemented by exceptional communication, leadership, and customer relationship management abilities. Passionate about leveraging technology to solve complex problems and drive business growth across multiple industries.

Technical Skills

Category	Technologies
Programming Languages	C, C++, C#, TypeScript, JavaScript, Python, Java, SQL, Bash
AI Tools & Agentic Development	Cline, Kilo, Claude Code, Google Antigravity, Github CoPilot, Perplexity, Figma, Google Stitch, MCP Servers & Agent Skills Development
Cloud Platforms & Services	Google Cloud Compute, Google Colab, Google Drive, Cloudflare Workers, Neon Serverless PostgreSQL, Supabase
Backend Frameworks	Node.js, Express.js, NestJS, Spring Boot 3.2, Django, Flask, FastAPI, REST APIs
DevOps & Infrastructure	Docker, Kubernetes, Terraform, CI/CD (GitHub Actions, GitLab CI), Nginx, Maven, Serverless Architecture
Databases & Caching	PostgreSQL (Neon, Supabase), SQL Server, MongoDB, Redis, SQLite, SQL Server Management Studio
Frontend Technologies	React 18, TypeScript, Vite, Tailwind CSS, Konva.js, JavaFX 17, HTML, CSS, JavaScript
AI/ML & Data Processing	PyTorch, Transformers, Granite 4.0 1B, Phi-3-mini, SQLite-vec, FTS5, sentence-transformers, NLP, Vector Embeddings
Message Queues & Streaming	RabbitMQ, Apache Kafka, Redis Pub/Sub
Architecture & Design Patterns	Microservices, Event-Driven Architecture, CQRS, Event Sourcing, Domain-Driven Design, SOLID Principles, 3NF Database Design
Security & Authentication	JWT, OAuth 2.0, RBAC, PCI Compliance, Authentication/Authorization Systems
Tools & Methodologies	Git, WebSocket/STOMP, Spring WebSocket, Hibernate, JPA, Gradio, Hugging Face, GitHub Actions, Docker Compose
Testing & Quality Assurance	JUnit (Java), Jest (Node.js), pytest (Python), Integration Testing, End-to-End Testing, Performance Testing, ESLint, Prettier, API Testing
Office Tools	Google Docs, Google Sheets, Google Slides, Microsoft Word, Excel, PowerPoint, Overleaf (LaTeX)
Cloud-Specific Services	Google Cloud Functions, AWS Lambda, Google Cloud Storage, S3, Cloudflare Workers, Serverless Architecture, Container Orchestration
Automation Tools	Zapier, n8n, GitHub Actions, GitLab CI, Jenkins, Workflow Automation, Task Scheduling

Professional Experience

Private Tutor (Independent Contractor) | Dhaka, Bangladesh | November 2023 – Present

- * Tutored 12 students in IGCSE/IAL Physics, Chemistry, Mathematics, and English, maintaining 7 active students
- * Delivered university-level instruction in SPL, OOP, Discrete Mathematics, and Probability & Statistics
- * Developed and leveraged Syllab AI platform with interactive chatbot interface for personalized tutoring and real-time feedback
- * Achieved A* grades in O-Levels and A grades in A-Levels through customized learning strategies and AI-powered tools
- * Created individualized lesson plans addressing learning gaps using both traditional and AI-driven educational technologies

SkyTech Solutions | Dhaka, Bangladesh | February 2023 – November 2023

Closer (Promoted from Telesales Specialist) | August 2023 – November 2023

- * Promoted after demonstrating exceptional telesales performance
- * Closed sales from telesales leads for government rebate heat pump programs
- * Developed advanced negotiation techniques to overcome customer objections
- * Mentored telesales team members to improve lead quality and conversion rates

Telesales Specialist | February 2023 – August 2023

- * Conducted outbound calls to Australian customers about government heat pump rebates
- * Built rapport by explaining complex rebate programs and financial benefits clearly
- * Maintained high call volume and conversion rates with Australian telecommunications compliance
- * Developed product knowledge and customer service skills through comprehensive training

CityGroup Dealership (Family Business) | Dhaka, Bangladesh | March 2018 – January 2023

Store Manager & Business Operations

- * Managed full retail operations including inventory management, stock receiving, and product display optimization
- * Oversaw financial operations: daily cash handling, mobile recharge services, mobile banking transactions, and comprehensive accounting
- * Supervised delivery team and field workers with proper goods tracking and accountability
- * Maintained detailed customer account records and outstanding payments
- * Operated independently as sole manager during peak periods handling all business operations
- * Developed expertise in multi-service retail combining traditional grocery sales with financial services and telecommunications

Key Projects

NexHire – High-Scale Recruitment Management System

 nexhire-frontend.vercel.app | **Tech:** SQL Server, CLR, C# .NET, T-SQL, Node.js, Express.js, React, Vite, Tailwind CSS, REST API, JWT, PostgreSQL, Neon, Redis

- * Designed normalized relational database schema following 3NF principles across 60+ entities with temporal data modeling, state machine formalization, and ACID compliance
- * Implemented string similarity algorithms for candidate matching: Jaro-Winkler, Levenshtein edit distance, and TF-IDF weighted cosine similarity achieving 94% precision
- * Developed predictive models: anti-ghosting risk scorer (0.78 AUC-ROC), onboarding success predictor (85% accuracy), and interviewer bias detection using statistical analysis
- * Built NLP pipeline for resume analysis: PDF/DOCX text extraction, named entity recognition, experience duration parsing, and sentiment analysis using VADER algorithm
- * Optimized performance: query execution plan analysis, covering indexes (60% cost reduction), connection pooling (70% latency reduction), and 10x speedup on aggregate queries

Syllab AI – Advanced learning platform (Ongoing)

 syllabai-frontend.vercel.app | **Tech:** Python, Transformers, PyTorch, Granite 4.0 1B, Phi-3-mini, SQLite, sqlite-vec, FTS5, sentence-transformers, Gradio, Hugging Face Spaces, Google Colab, FastAPI, HTML, CSS, JavaScript, JSONL, PEFT, LoRA, GGUF

- * Developed hybrid AI system combining fine-tuned open-source language models (Granite 4.0 1B, Phi-3-mini) with SQLite vector database for RAG architecture enabling accurate, context-aware student responses
- * Designed comprehensive SQLite database schema with FTS5 and vector embeddings storing 500+ past paper questions with metadata across Edexcel and Cambridge IGCSE/IAL Physics
- * Fine-tuned multiple open-source language models on 1000+ curated question-answer pairs achieving 40% faster inference with GGUF quantization while maintaining 95% accuracy
- * Built interactive chatbot interface using Gradio on Hugging Face Spaces with efficient model loading and sub-3-second response times on free-tier GPU infrastructure
- * Designed modular platform architecture supporting multiple curricula with past paper analysis, topic extraction, quizzes, and AI tutoring with database-agnostic design for cloud migration

Unnamed Multiplayer Game/Game Engine Project - (Ongoing)

 github.com/nawaf-al-hussain | **Tech:** Java 17, Spring Boot 3.2, Spring WebSocket, STOMP, Redis, PostgreSQL (Neon), Hibernate, JPA, Maven, Docker, React 18, TypeScript, Vite, Konva.js, Tailwind CSS, JavaFX 17, Scene Builder, JWT, WebSocket, REST APIs, GitHub Actions

- * Designed universal, event-driven board game engine in Java 17 using DDD and event sourcing principles, powering multiple games (The Game of Life 2, Monopoly, Clue) from single modular core with zero game-specific coupling
- * Architected distributed, server-authoritative multiplayer system using Spring Boot 3 WebSocket (STOMP) and Redis Pub/Sub enabling real-time cross-platform gameplay with horizontal scaling and live state synchronization
- * Built complete event-sourced persistence layer with PostgreSQL (Neon) for immutable event storage including snapshot management, full game replayability, and CQRS-inspired command → event → state pipeline
- * Developed dual frontend clients: React 18 + Konva.js web application and JavaFX 17 desktop client, both communicating via WebSocket and REST to shared, authoritative game server
- * Engineered extensible plugin system allowing third-party game logic (stock market expansions, AI players, disaster mechanics) to be dynamically injected without modifying core code

Education

Bachelor of Science in Computer Science & Engineering United International University, Dhaka, Bangladesh | June 2024 – Present

IGCSE British Council – Private Candidate, Dhaka, Bangladesh | May/June 2019 – January 2020

- * Relevant subjects: Mathematics B, Physics, Chemistry, English, ICT, Bangla
- * Grades: A*, A*, A*, A*, A, A*
- * Achievements: Distinctions in Mathematics B and Physics

IAL (International Advanced Levels) British Council – Private Candidate, Dhaka, Bangladesh | May/June 2022 – May/June 2023

- * Relevant subjects: Pure Mathematics, Physics
- * Grades: A, A
- * Achievements: Full UMS marks in all AS level units

Additional Information

- * **Languages:** Bengali (Native), English (Professional Working Proficiency)
- * **Open Source:** Active contributor to Node.js ecosystem projects on GitHub
- * **Interests:** Distributed Systems, Cloud Architecture, AI/ML Integration, Developer Experience